

Lecture 18, Mar 09 2026

Example. a). Find the tangent to the cycloid $x = r(\theta - \sin \theta)$

$$y = r(1 - \cos \theta) \quad \text{at } \theta = \pi/3.$$

b). At what points is the tangent horizontal? When is it vertical?

sln. a). ✓

b). We know that

$$\frac{dy}{dx} = \frac{r \sin \theta}{r(1 - \cos \theta)} = \frac{\sin \theta}{1 - \cos \theta}.$$

the tangent line is horizontal when $\frac{dy}{dx} = 0 \Leftrightarrow \sin \theta = 0$ and

$1 - \cos \theta \neq 0$ then $\theta = (2n-1)\pi$, $n \in \mathbb{Z}$. The corresponding point on the cycloid is $((2n-1)\pi r, 2r)$.

when $\theta = 2n\pi$, $n \in \mathbb{Z}$ both $dx/d\theta$ and $dy/d\theta$ are 0.

We need to check that at $\theta = 2n\pi$ we have vertical tangent lines.

$$\lim_{\theta \rightarrow 2n\pi^+} \frac{dy}{dx} = \lim_{\theta \rightarrow 2n\pi^+} \frac{\sin \theta}{1 - \cos \theta} \stackrel{(\frac{0}{0})}{=} \lim_{\theta \rightarrow 2n\pi^+} \frac{\cos \theta}{\sin \theta} = \infty.$$

Similarly $\lim_{\theta \rightarrow 2n\pi^-} \frac{dy}{dx} = -\infty$ so, indeed there are vertical tangents when $\theta = 2n\pi$ and the points are $(2n\pi r, 0)$.

Arc Length. Let consider $y = f(x)$ for $a \leq x \leq b$. The length of the curve $y = f(x)$ is

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

Thm. If C is described by the parametric equations

$$C: \begin{cases} x = f(t) \\ y = g(t) \end{cases} \quad \alpha \leq t \leq \beta.$$

where, f', g' are continuous. Then, the length of C is

$$L = \int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

Example: let consider a circle of radius $r=1$.

Compute its perimeter.

Sln. We know that $x^2 + y^2 = 1$. So,

$$x = \cos \theta$$

$$0 \leq \theta \leq 2\pi.$$

$$y = \sin \theta$$

$$\frac{dx}{d\theta} = -\sin \theta, \quad \frac{dy}{d\theta} = \cos \theta.$$

$$\begin{aligned} L &= \int_0^{2\pi} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta = \int_0^{2\pi} \sqrt{(-\sin \theta)^2 + \cos^2 \theta} d\theta \\ &= \int_0^{2\pi} \sqrt{\sin^2 \theta + \cos^2 \theta} d\theta = \int_0^{2\pi} d\theta = 2\pi. \end{aligned}$$

$$P = 2\pi r. \quad r=1 \quad \Rightarrow \quad P = 2\pi.$$

Example. Find the length of one arch of the cycloid

$$x = r(\theta - \sin \theta)$$

$$0 \leq \theta \leq 2\pi.$$

$$y = r(1 - \cos \theta).$$

Sln.
$$\frac{dx}{d\theta} = \frac{d}{d\theta} (r(\theta - \sin \theta)) = r \frac{d}{d\theta} (\theta - \sin \theta) = r(1 - \cos \theta)$$

$$\frac{dy}{d\theta} = r \sin \theta$$

$$\begin{aligned}
L &= \int_0^{2\pi} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta = \int_0^{2\pi} \sqrt{(r(1-\cos\theta))^2 + (r\sin\theta)^2} d\theta \\
&= \int_0^{2\pi} \sqrt{r^2 - 2r^2\cos\theta + \underbrace{r^2\cos^2\theta + r^2\sin^2\theta}_{r^2}} d\theta \\
&= \int_0^{2\pi} \sqrt{r^2(2 - 2\cos\theta)} d\theta \\
&= \int_0^{2\pi} r \sqrt{2(1-\cos\theta)} d\theta \\
&= r \int_0^{2\pi} \sqrt{4\sin^2(\theta/2)} d\theta \\
&= 2r \int_0^{2\pi} \sin(\theta/2) d\theta \\
&= 2r \left\{ -2\cos(\theta/2) \right\}_0^{2\pi} = 2r \{2 + 2\} = 8r.
\end{aligned}$$

We know $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$

So, $\sin^2(\theta/2) = \frac{1}{2}(1 - \cos\theta)$

$2\sin^2(\theta/2) = (1 - \cos\theta)$

Example. The position of a particle at time t is given

$$x = 2t + 3$$

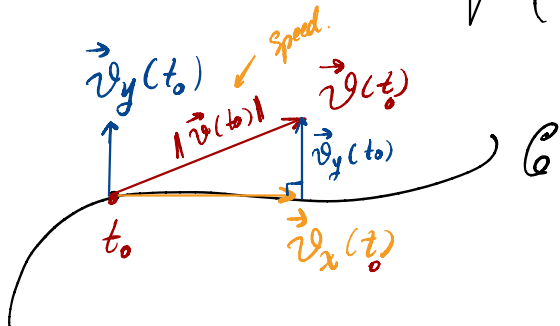
$$y = 4t^2, \quad t \geq 0.$$

a). Eliminate the parameter.

b). Find the speed of the particle when it is at the point (5, 4).

sol. Def.

$$v(t) = s'(t) = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$



$$\begin{aligned}
\|\vec{v}(t_0)\| &= \sqrt{(v_x(t_0))^2 + (v_y(t_0))^2} \\
&= \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}
\end{aligned}$$

$$\frac{dx}{dt} = 2 \quad \frac{dy}{dt} = 8t$$

then $v(t) = \sqrt{2^2 + (8t)^2} = 2\sqrt{1 + 16t^2}$

We want $(5, 4)$ then

$$\begin{cases} 5 = 2t + 3 \\ 4 = 4t^2 \end{cases} \Rightarrow t = 1.$$

$$v(1) = 2\sqrt{1 + 16} = 2\sqrt{17}$$